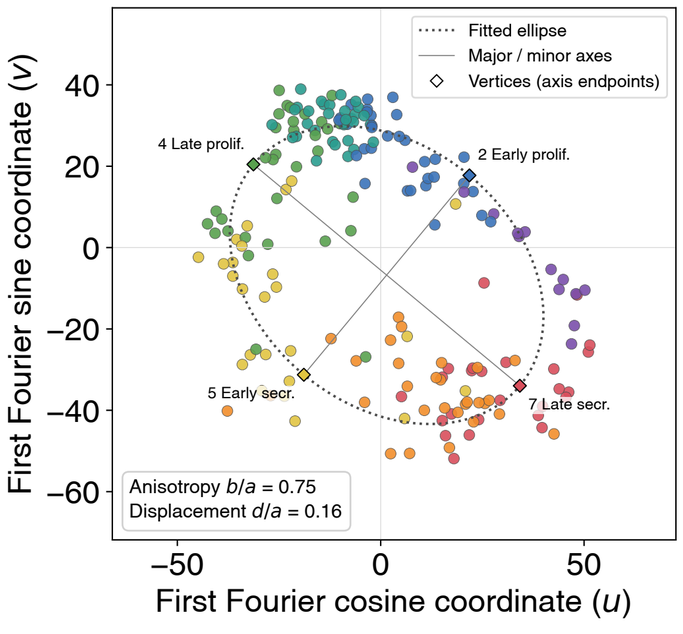
A

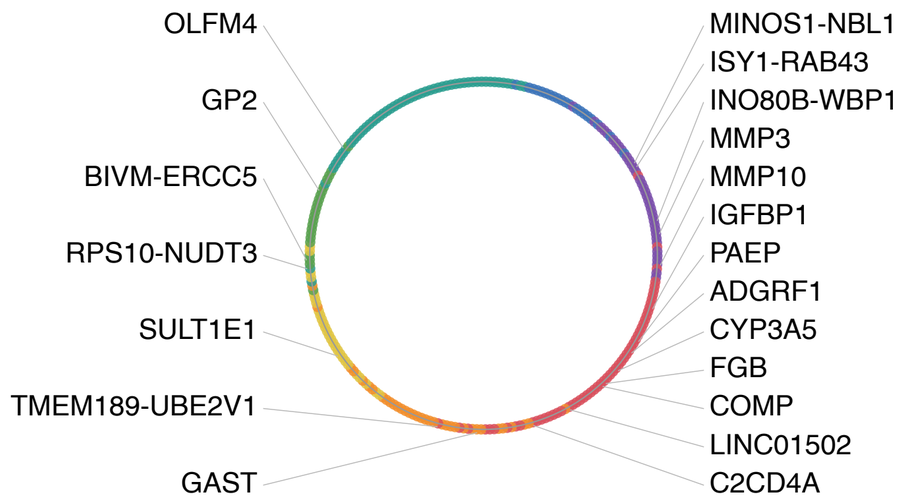
Histological stage

$\rho_{\text{circ cycle}}$=0.841, $R_{\text{histology}}$=0.867



B

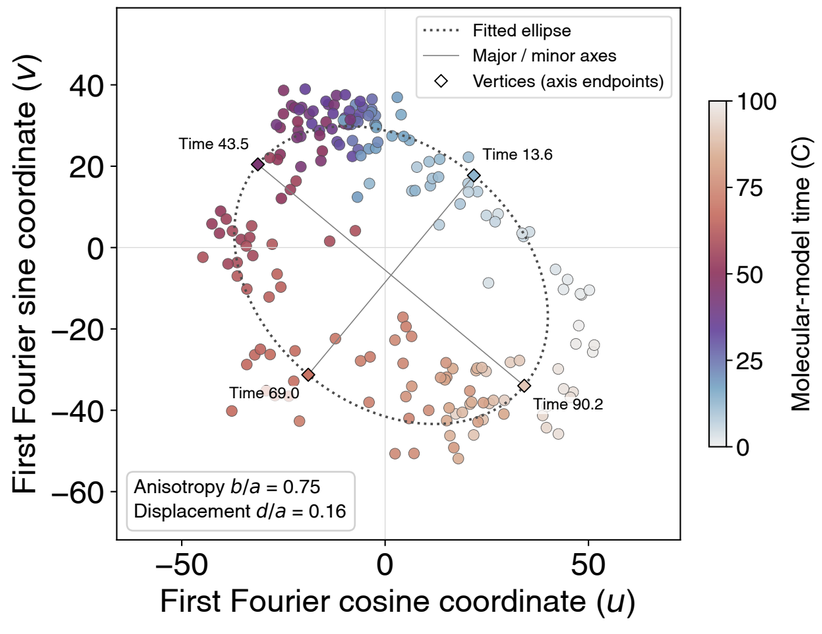
Learned gene circle



C

Authors' molecular-model time

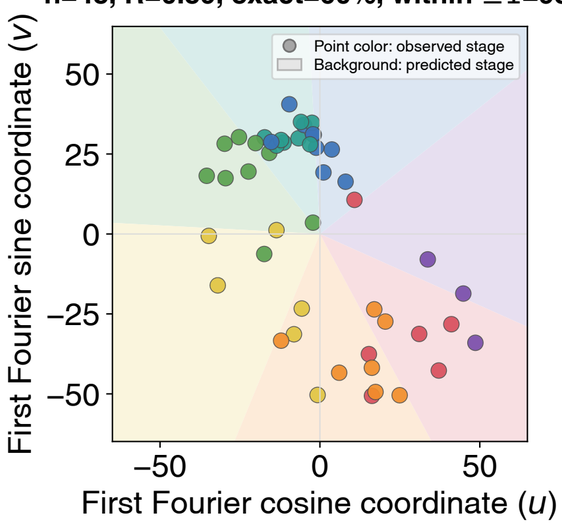
$\rho_{\text{circ cycle}}$=0.841, $R_{\text{molecular}}$=0.922



D

GSE234354 held-out

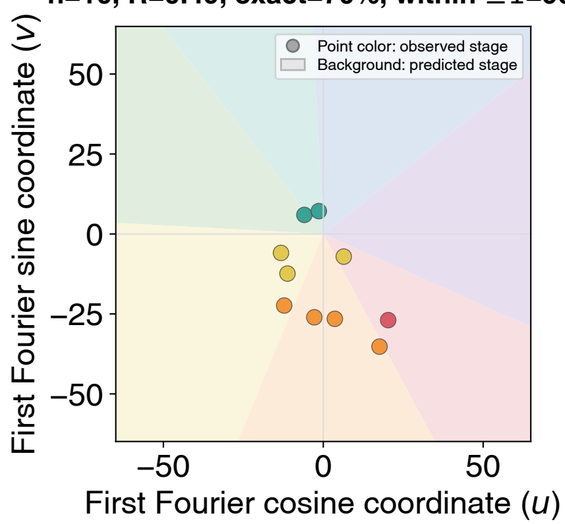
n=48, R=0.89, exact=60%, within ±1=98%



E

GSE111976 10x pseudobulk

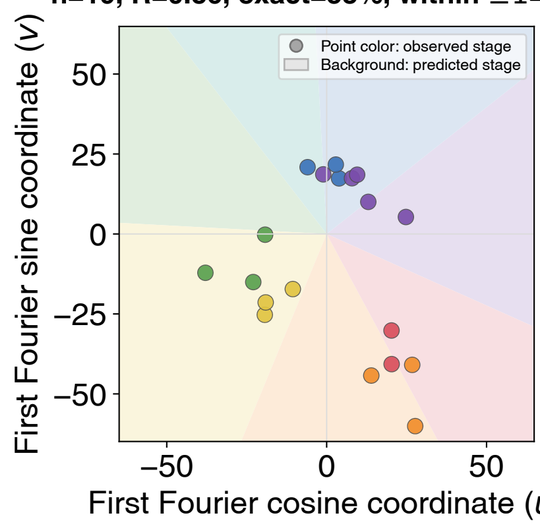
n=10, R=0.49, exact=70%, within ±1=90%



F

GSE111976 C1 pseudobulk

n=19, R=0.86, exact=53%, within ±1=95%



Unified histological-stage colors shared by A, B, and D–F

1 Menstrual · 2 Early proliferative · 3 Mid proliferative · 4 Late proliferative · 5 Early secretory · 6 Mid secretory · 7 Late secretory